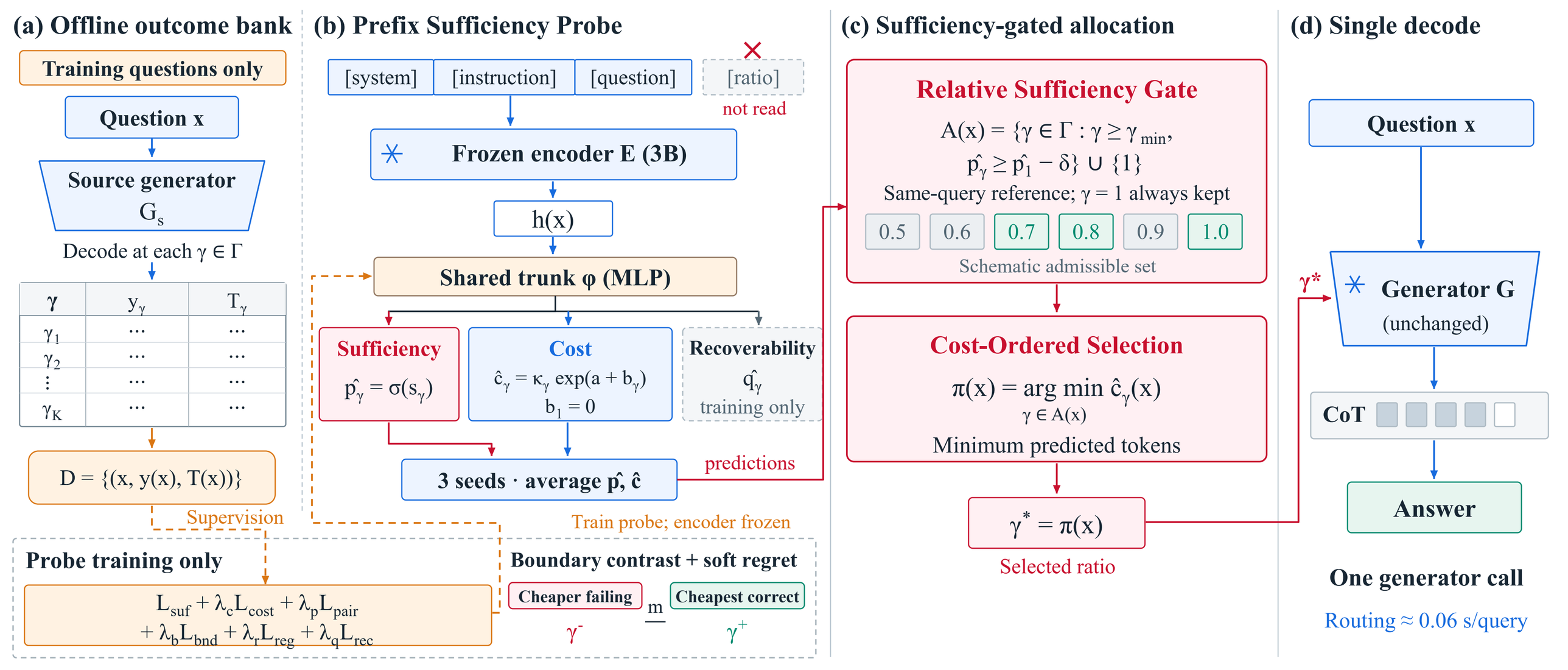

(a) Offline outcome bank
Training questions only
Question x
Source generator
G_s
Decode at each $\gamma \in \Gamma$
γ | y_γ | T_γ
γ_1 | … | …
γ_2 | … | …
⋮ | … | …
γ_K | … | …
$D = \{(x, y(x), T(x))\}$
Supervision
Probe training only
$L_{suf} + \lambda_c L_{cost} + \lambda_p L_{pair} + \lambda_b L_{bnd} + \lambda_r L_{reg} + \lambda_q L_{rec}$
(b) Prefix Sufficiency Probe
[system] [instruction] [question]
[ratio]
not read
Frozen encoder E (3B)
h(x)
Shared trunk φ (MLP)
Sufficiency
$\hat{p}_\gamma = \sigma(s_\gamma)$
Cost
$\hat{c}_\gamma = \kappa_\gamma \exp(a + b_\gamma)$
$b_1 = 0$
Recoverability
$\hat{q}_\gamma$
training only
3 seeds · average $\hat{p}$, $\hat{c}$
predictions
Train probe; encoder frozen
Boundary contrast + soft regret
Cheaper failing
m
Cheapest correct
γ^-
γ^+
(c) Sufficiency-gated allocation
Relative Sufficiency Gate
$A(x) = \{\gamma \in \Gamma : \gamma \geq \gamma_{min}, \hat{p}_\gamma \geq \hat{p}_1 - \delta\} \cup \{1\}$
Same-query reference; $\gamma = 1$ always kept
0.5 0.6 0.7 0.8 0.9 1.0
Schematic admissible set
Cost-Ordered Selection
$\pi(x) = \arg\min \hat{c}_\gamma(x)$
$\gamma \in A(x)$
Minimum predicted tokens
$\gamma^* = \pi(x)$
Selected ratio
(d) Single decode
Question x
γ^*
Generator G
(unchanged)
CoT
Answer
One generator call
Routing ≈ 0.06 s/query